

Current Electricity

Questions with Answer Keys

JEE Main 2020 Chapterwise

Physics

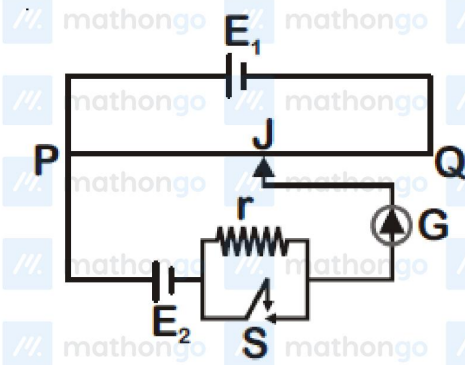
Q1 JEE Main 2020 - 2 September (Morning)

Consider four conducting materials copper, tungsten, mercury and aluminium with resistivity ρ_C , ρ_T , ρ_M and ρ_A respectively

- (A) $\rho_M > \rho_A > \rho_C$
- (B) $\rho_C > \rho_A > \rho_T$
- (C) $\rho_A > \rho_M > \rho_C$
- (D) $\rho_A > \rho_T > \rho_C$

Q2 JEE Main 2020 - 2 September (Evening)

A potentiometer wire PQ of 1m length is connected to a standard cell E_1 . Another cell E_2 of emf 1.02V is connected with a resistance 'r' and switch S (as shown in figure). With switch S open, the null position is obtained at a distance of 49cm from Q. The potential gradient in the potentiometer wire is



- (A) 0.04V/cm
- (B) 0.01V/cm
- (C) 0.02V/cm
- (D) 0.03V/cm

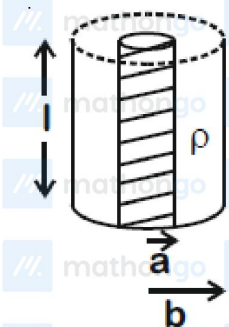
Q3 JEE Main 2020 - 2 September (Evening)

An ideal cell of emf 10V is connected in circuit shown in figure. Each resistance is 2Ω . The potential difference (in V) across the capacitor when it is fully charged is



Q4 JEE Main 2020 - 3 September (Morning)

Model a torch battery of length l to be made up of a thin cylindrical bar of radius ' a ' and a concentric thin cylindrical shell of radius ' b ' filled in between with an electrolyte of resistivity ρ (see figure). If the battery is connected to a resistance of value R , the maximum Joule heating in R will take place for



- (A) $R = \frac{\rho}{\pi d} \ln\left(\frac{b}{a}\right)$
 (B) $R = \frac{2\rho}{\pi d} \ln\left(\frac{b}{a}\right)$
 (C) $R = \frac{\rho}{2\pi d} \ln\left(\frac{b}{a}\right)$
 (D) $R = \frac{\rho}{2\pi} \left(\frac{b}{a}\right)$

Q5 JEE Main 2020 - 3 September (Evening)

Which of the following will NOT be observed when a multimeter (operating in resistance measuring mode) probes connected across a component, are just reversed?

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Physics

- (A) Multimeter shows NO deflection in both cases i.e. before and after reversing the probes if the chosen component is capacitor.
- (B) Multimeter shows NO deflection in both cases i.e. before and after reversing the probes if the chosen component is metal wire.
- (C) Multimeter shows a deflection, accompanied by a splash of light out of connected component in one direction and NO deflection on reversing the probes if the chosen component is LED.
- (D) Multimeter shows an equal deflection in both cases i.e. before and after reversing the probes if the chosen component is resistor.

Q6 JEE Main 2020 - 3 September (Evening)

Two resistors 400Ω and 800Ω are connected in series across a $6V$ battery. The potential difference measured by a voltmeter of $10k\Omega$ across 400Ω resistor is close to

- (A) $2.05V$
(B) $1.8V$
(C) $2V$
(D) $1.95V$

Q7 JEE Main 2020 - 3 September (Evening)

A galvanometer coil has 500 turns and each turn has an average area of $3 \times 10^{-4}m^2$. If a torque of $1.5Nm$ is required to keep this coil parallel to a magnetic field when a current of $0.5A$ is flowing through it, the strength of the field (in T) is _____.

Q8 JEE Main 2020 - 4 September (Morning)

A battery of $3.0V$ is connected to a resistor dissipating $0.5W$ of power. If the terminal voltage of the battery is $2.5V$, the power dissipated within the internal resistance is

- (A) $0.10W$
(B) $0.072W$
(C) $0.50W$
(D) $0.125W$

Q9 JEE Main 2020 - 4 September (Evening)

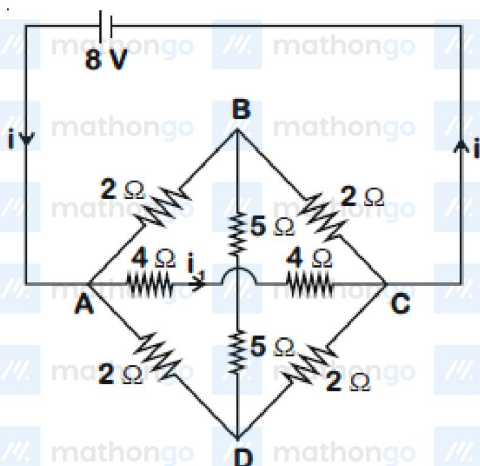
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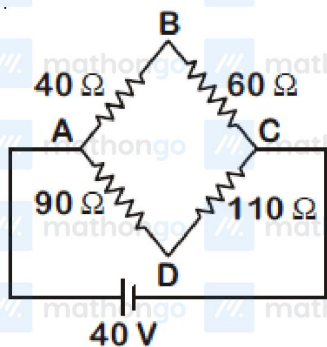
Physics

The value of current i_1 flowing from A to C in the circuit diagram is :



- (A) 1A
- (B) 4A
- (C) 5A
- (D) 2A

Q10 JEE Main 2020 - 4 September (Evening)



Four resistances 40Ω , 60Ω , 90Ω and 110Ω make the arms of a quadrilateral ABCD. Across AC is a battery of emf $40V$ and internal resistance negligible. The potential difference across BD in V is

Q11 JEE Main 2020 - 5 September (Morning)

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Physics

An electrical power line, having a total resistance of 2Ω , delivers 1kW at 220V . The efficiency of the transmission line is approximately

- (A) 85%
- (B) 96%
- (C) 72%
- (D) 91%

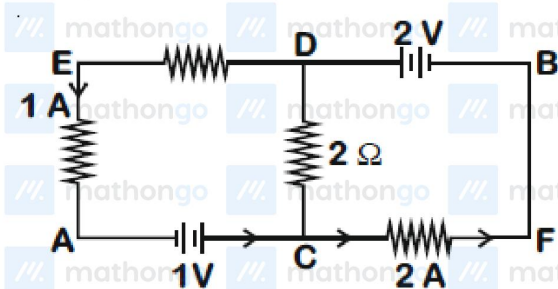
Q12 JEE Main 2020 - 5 September (Morning)

A galvanometer of resistance G is converted into a voltmeter of range $0 - 1\text{V}$ by connecting a resistance R_1 in series with it. The additional resistance that should be connected in series with R_1 to increase the range of the voltmeter to $0 - 2\text{V}$ will be

- (A) G
- (B) R_1
- (C) $R_1 + G$
- (D) $R_1 - G$

Q13 JEE Main 2020 - 5 September (Evening)

In the circuit, given in the figure currents in different branches and value of one resistor are shown. Then potential at point B with respect to the point A is



- (A) $+1\text{V}$
- (B) -2V
- (C) $+2\text{V}$
- (D) -1V

Q14 JEE Main 2020 - 5 September (Evening)

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Physics

A galvanometer is used in laboratory for detecting the null point in electrical experiments. If, on passing a current of 6mA it produces a deflection of 2° , its figure of merit is close to

(A) $6 \times 10^{-3} \text{ A/div.}$

(B) 666° A/div.

(C) $3 \times 10^{-3} \text{ A/div.}$

(D) 333° A/div.

Q15 JEE Main 2020 - 6 September (Evening)

A circuit to verify Ohm's law uses ammeter and voltmeter in series or parallel connected correctly to the resistor. In the circuit

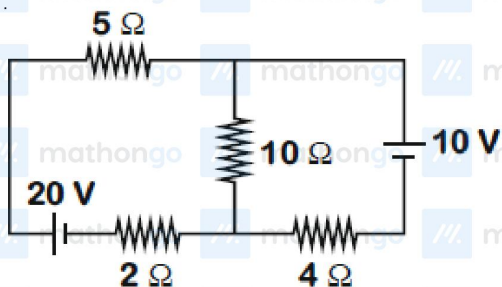
(A) Ammeter is always connected in series and voltmeter in parallel

(B) Both ammeter and voltmeter must be connected in parallel

(C) Ammeter is always used in parallel and voltmeter is series

(D) Both ammeter and voltmeter must be connected in series

Q16 JEE Main 2020 - 6 September (Evening)



In the figure shown, the current in the 10V battery is close to

(A) 0.36 A from negative to positive terminal

(B) 0.42 A from positive to negative terminal

(C) 0.71 A from positive to negative terminal

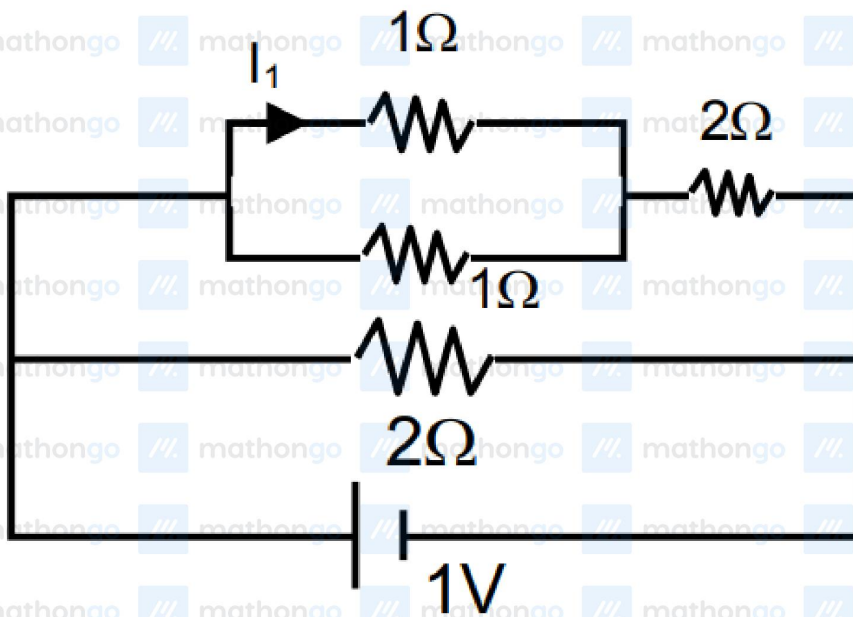
(D) 0.21 A from positive to negative terminal

Q17 JEE Main 2020 - 7 January (Morning)

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The current ' i_1 ' in the given circuit is



- (A) 0.2 A
- (B) 0.3 A
- (C) 0.5 A
- (D) 0.25 A

Q18 JEE Main 2020 - 7 January (Evening)

In a building there are 15 bulbs of 45 W, 15 bulbs of 100 W, 15 small fans of 10 W and 2 heaters of 1 kW. The voltage of electric main is 220 V. The minimum fuse capacity (rated value) of building will be:

- (A) 5 A
- (B) 20 A
- (C) 25 A
- (D) 15 A

Q19 JEE Main 2020 - 7 January (Evening)

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A battery of unknown emf connected to a potentiometer has balancing length 560 cm . If a resistor of resistance $10\ \Omega$ is connected in parallel with the cell the balancing length change by 60 cm . If the internal resistance of the cell is $\frac{n}{10}\ \Omega$, the value of 'n' is

Q20 JEE Main 2020 - 8 January (Morning)

The length of a potentiometer wire is 120 cm and it carries a current of 60 mA . For a cell of emf 5 V and internal resistance of $20\ \Omega$, the null point on it is found to be at 100 cm . The resistance of whole wire is:

- (A) $80\ \Omega$
- (B) $100\ \Omega$
- (C) $120\ \Omega$
- (D) $60\ \Omega$

Q21 JEE Main 2020 - 8 January (Morning)

Four resistors of $15\ \Omega$, $4\ \Omega$ and $10\ \Omega$ are arranged in cyclic order to form a Wheatstone bridge. What resistance (in Ω) should be connected in parallel across the $10\ \Omega$ resistor to balance the Wheatstone bridge.

Q22 JEE Main 2020 - 8 January (Evening)

A galvanometer having a coil resistance $100\ \Omega$ gives a full scale deflection when a current of 1 mA is passed through it. What is the value of the resistance which can convert this galvanometer into a voltmeter giving full scale deflection for a potential difference of 10 V ?

- (A) $0.99\text{ K}\Omega$
- (B) $9.9\text{ K}\Omega$
- (C) $9.8\text{ K}\Omega$
- (D) $10\text{ K}\Omega$

Q23 JEE Main 2020 - 8 January (Evening)

The series combination of two batteries, both of the same emf 10 V , but different internal resistance of $20\ \Omega$ and $5\ \Omega$, is connected to the parallel combination of two resistors $30\ \Omega$ and $R\ \Omega$. The voltage difference across the battery of internal resistance $20\ \Omega$ is zero, the value of R (in Ω) is

Current Electricity

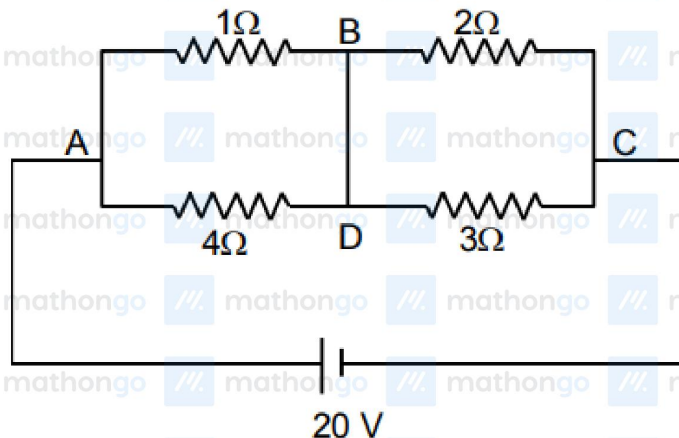
Questions with Answer Keys

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Physics

Q24 JEE Main 2020 - 9 January (Morning)

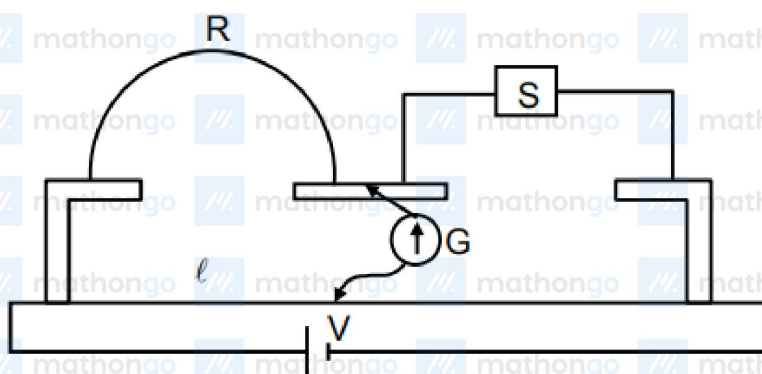
In the given circuit diagram, a wire is joining points B and D . The current in this wire is :



- (A) 1.6 A
- (B) 2 A
- (C) 2.4 A
- (D) 3 A

Q25 JEE Main 2020 - 9 January (Evening)

In a meter bridge experiment S is a standard resistance. R is a resistance wire. It is found that balancing length is $\ell = 25 \text{ cm}$. If R is replaced by a wire of half length and half diameter that of R of same material, then the balancing distance ℓ' (in cm) will now be



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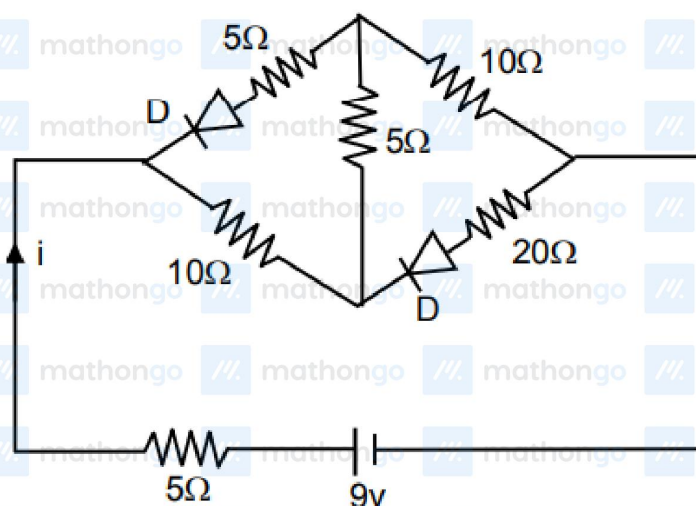
Current Electricity

Questions with Answer Keys

JEE Main 2020 Chapterwise

Physics

Q26 JEE Main 2020 - 9 January (Evening)



The current i in the network is :

- (A) 0.2 A
- (B) 0.3 A
- (C) 0 A
- (D) 0.6 A

Current Electricity

Questions with Answer Keys

JEE Main 2020 Chapterwise

Physics

Answer Key

Q1 (A)

Q2 (C)

Q3 (8)

Q4 (C)

Q5 (A)

Q6 (D)

Q7 (20)

Q8 (A)

Q9 (A)

Q10 (2)

Q11 (B)

Q12 (C)

Q13 (A)

Q14 (C)

Q15 (A)

Q16 (D)

Q17 (A)

Q18 (B)

Q19 (12)

Q20 (B)

Q21 (10)

Q22 (B)

Q23 (30)

Q24 (B)

Q25 (40)

Q26 (B)

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Current Electricity

Hints & Solutions

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Physics

Q1

$$\rho_M = 98 \times 10^{-8}$$

$$\rho_A = 2.80 \times 10^{-8}$$

$$\rho_C = 1.72 \times 10^{-8}$$

$$\rho_T = 5.65 \times 10^{-8}$$

$$\rho_M > \rho_T > \rho_A > \rho_C$$

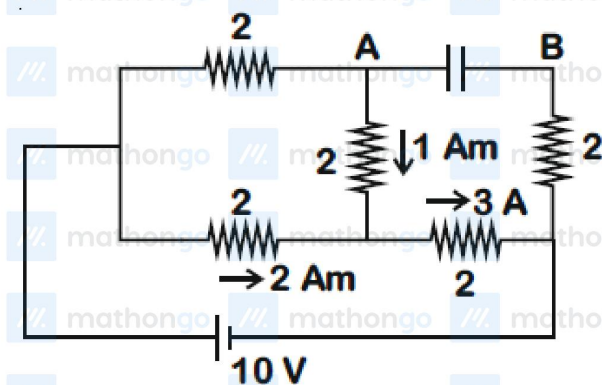
Q2

$$\text{Potential gradient} = \frac{\text{Potential drop}}{\text{length}}$$

$$= \frac{1.02}{100 - 49} \frac{\text{V}}{\text{cm}}$$

$$= 0.02 \frac{\text{V}}{\text{cm}}$$

Q3



After solving the circuit we got the final current distribution as shown in the above diagram. So potential difference between A and B is

$$0 + (2 \times 3) + (2 \times 1) = 8 \text{ volt}$$

Q4

$$\text{Power in external } R = \left(\frac{\epsilon}{r+R} \right)^2 R$$

This power is maximum, when $R = r$

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Hints & Solutions

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Physics

$$r = \int_a^b \rho \frac{dx}{2\pi x}$$

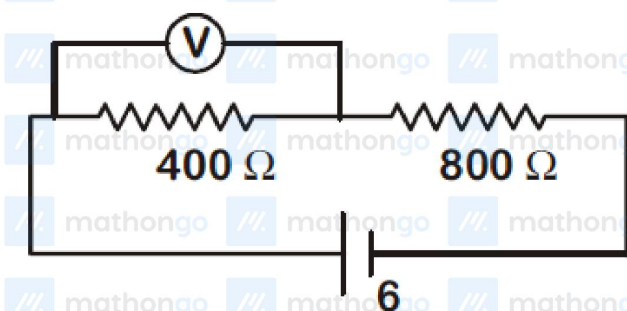
$$r = \frac{\rho}{2\pi l} \ln \frac{b}{a}$$

Q5

Theory based

Q6

$$\text{Voltmeter Reading} = \frac{V \times 400R}{400+R}$$



$$I = \frac{6}{\frac{400R}{400+R} + 800}$$

Putting the value of R, we get

$$\text{Reading} = 1.95 \text{ V}$$

Q7

$$\vec{\mu} = NiA = 500 \times 3 \times 10^{-4} \times \frac{5}{10}$$

$$\tau = \vec{\mu} \times \vec{\beta} = \mu\beta \sin \theta \quad (\because \sin \theta = 1)$$

$$\therefore B = \frac{15 \times 10}{10 \times 500 \times 3 \times 5 \times 10^{-4}}$$

$$B = 20 \text{ T}$$

Q8

$$i^2 R = 0.5$$

$$iR = 2.5$$

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Hints & Solutions

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Physics



$$\Rightarrow i = 0.2 \text{ A} \text{ and } R = 12.5 \Omega$$

$$V = \varepsilon - ir$$

$$2.5 = 3.0 - 0.2r$$

$$\Rightarrow r = 2.5 \Omega$$

Power dissipated in internal resistance

$$= i^2 r = 0.1 \text{ W}$$

q9

$$\varepsilon - i_1 R - i_1 R = 0$$

$$8 - 4i_1 - 4i_1 = 0$$

$$i_1 = 1 \text{ A}$$

q10

$$V_B = \frac{40 \times 60}{100} = 24 \text{ volt}$$

$$V_D = \frac{40 \times 110}{200} = 22 \text{ volt}$$

$$\therefore \Delta V = V_B - V_D = 2 \text{ volt}$$

q11

$$I = \frac{P}{V} = \frac{(1000)}{220}$$

$$P_{\text{Loss}} = I^2 R_{\text{trans}} = \left(\frac{1000}{220} \right)^2 \times 2$$

$$\Rightarrow \text{efficiency} = \left(\frac{1000}{1000 + P_{\text{Loss}}} \right) \times 100$$

$$\simeq 96\%$$

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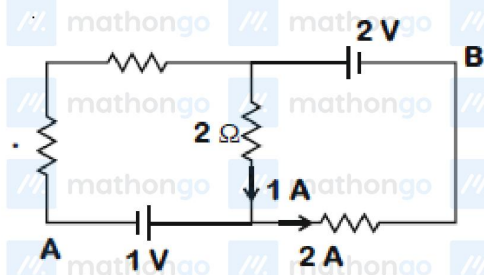
Physics

Q12

$$i_g (G + R_1) = 1 \quad i_g (G + R_1 + R') = 2$$

$$\Rightarrow R' = G + R_1$$

Q13



$$v_B + 2 - 2 \times 1 - 1 = V_A$$

$$\Rightarrow V_B - V_A = 3 - 2 = 1V$$

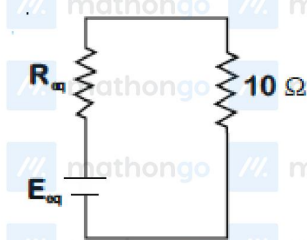
Q14

$$\text{Figure of merit} = \frac{\text{Current}}{\text{Deflection}}$$

Q15

Ammeter is always connected in series and voltmeter is connected in parallel.

Q16



$$\frac{1}{R_{eq}} = \frac{1}{4} + \frac{1}{7} \Rightarrow R_{eq} = \frac{28}{11} \Omega$$

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Hints & Solutions

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Physics

Circuit can be reduced to

$$\frac{11E_{eq}}{28} = \frac{20}{7} + \frac{10}{4}$$

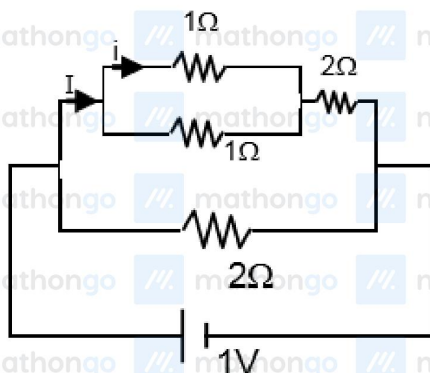
$$E_{eq} = \frac{150}{11} \text{ V}$$

$$I = \frac{\frac{150}{11}}{\frac{28}{11} + 10} = \frac{150}{138} \text{ A}$$

$$\frac{1500}{138} = 10 + 41$$

$$1 = \frac{30}{138} \text{ A}$$

Q17



$$I = \frac{1}{2.5} = 0.4 \text{ A}$$

$$i = \frac{I}{2} = 0.2 \text{ A}$$

Q18

$$\text{Total power is } (15 \times 45) + (15 \times 100) + (15 \times 10) + (2 \times 1000) \\ = 4325 \text{ W}$$

$$\text{So current is } = \frac{4325}{220} = 19.66 \text{ A}$$

$$= 20 \text{ Amp.}$$

Q19

Let the emf of cell is ε internal resistance is r and potential gradient is x .

only cell connected:

$$\varepsilon = 560x \quad \dots (1)$$

After connecting the resistor

$$\frac{\varepsilon \times 10}{10 + r} = 500x \quad \dots (2)$$

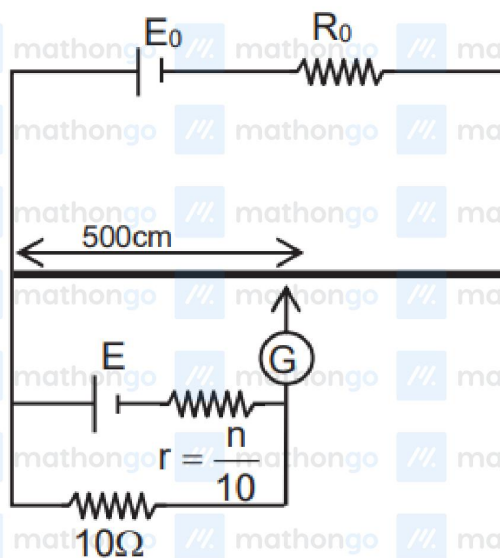
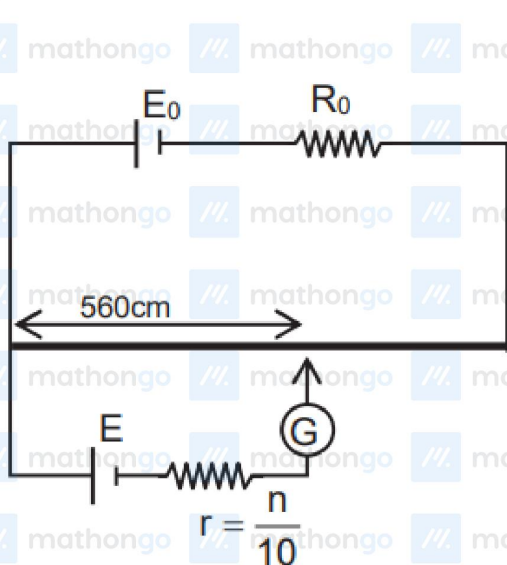
from (1) and (2)

$$\frac{560 \times 10}{10 + r} = 500x$$

$$56 = 50 + 5r$$

$$r = \frac{6}{5} = 1.2 \Omega$$

$$n = 12$$



Q20

Current Electricity

Hints & Solutions

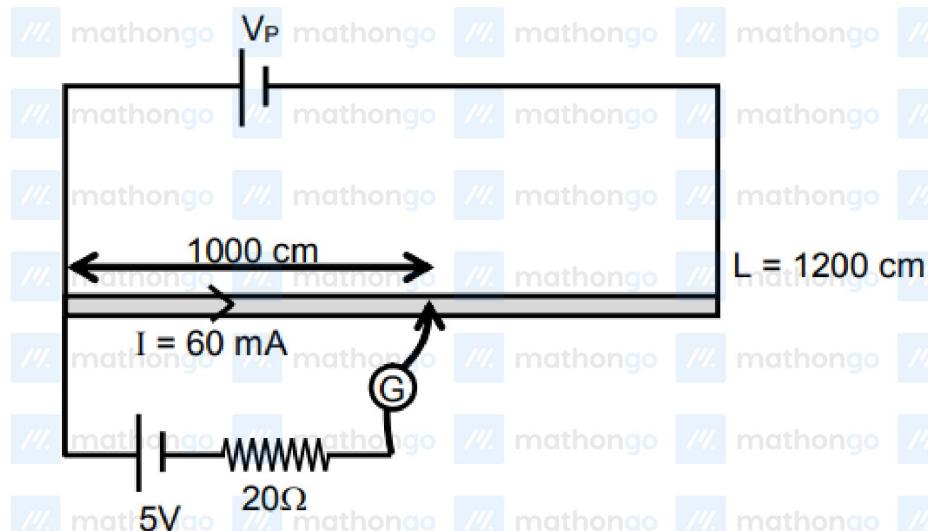
JEE Main 2020 Chapterwise

Physics

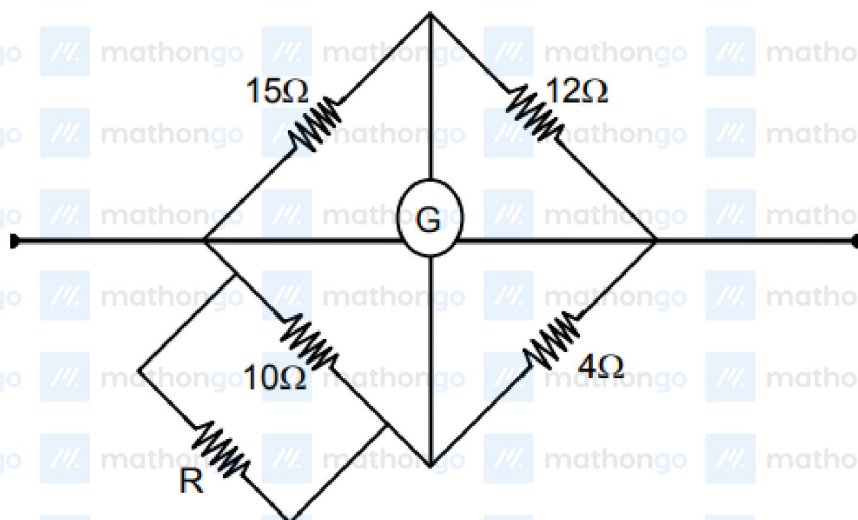
$$\text{Potential gradient} = \frac{5}{1000} = \frac{V_P}{1200}$$

$$V_P = 6V$$

$$\text{and } R_P = \frac{V_P}{I} = \frac{6}{60 \times 10^{-3}} = 100 \Omega$$



Q21



$$\frac{10R}{10+R} \times 12 = 15 \times 4$$

on solving

$$R = 10 \, \Omega$$

Q22

$$V_g = i_g R_g = 0.1 \, \text{V}$$

$$V = 10 \, \text{V}$$

$$R = R_g \left(\frac{V}{V_g} - 1 \right)$$

$$= 100 \times 99 = 9.9 \, \text{K}\Omega$$

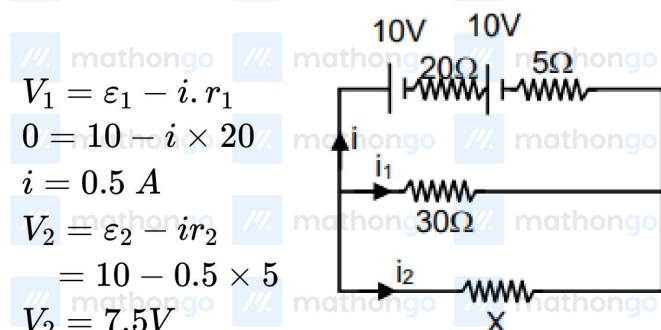
Q23

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Physics



$$V_1 = \varepsilon_1 - i \cdot r_1$$

$$0 = 10 - i \times 20$$

$$i = 0.5 \text{ A}$$

$$V_2 = \varepsilon_2 - i r_2$$

$$= 10 - 0.5 \times 5$$

$$V_2 = 7.5 \text{ V}$$

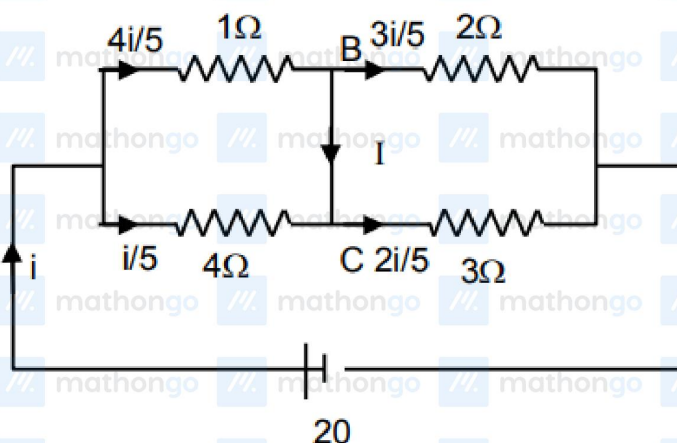
$$0.5 = \frac{7.5}{30} + \frac{7.5}{x}$$

$$0.5 = 0.25 + \frac{7.5}{x}$$

$$\frac{7.5}{x} = 0.25$$

$$x = \frac{7.5}{0.25} = 30$$

Q24



$$R_{eff} = \frac{4}{5} + \frac{6}{5} = 2 \Omega$$

$$i = \frac{20}{2}$$

$$= 10 \text{ A}$$

$$I = \frac{4i}{5} - \frac{3i}{5} = +\frac{i}{5} = 2 \text{ A}$$

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Q25

$$\frac{X}{R} = \frac{75}{25} = 3$$

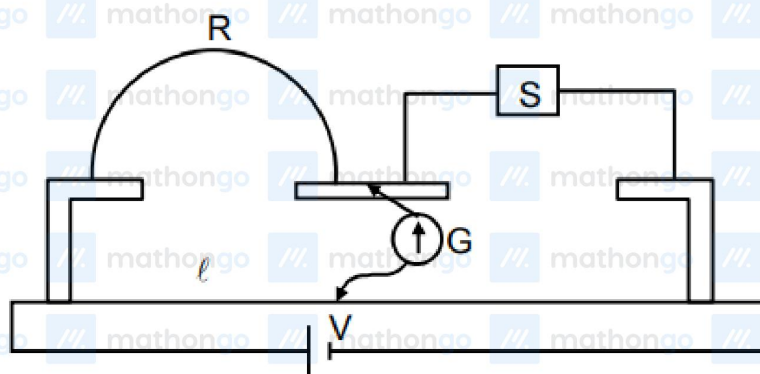
$$R = \frac{\rho \ell}{A} = \frac{4\rho \ell}{\pi d^2}$$

$$R' = \frac{4\rho \left(\frac{\ell}{2}\right)}{\pi \left(\frac{d}{2}\right)^2} = 2R$$

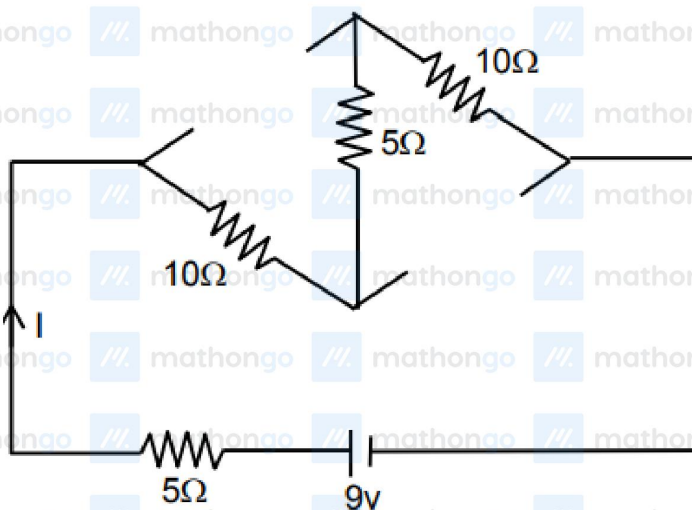
$$\text{then } \frac{X}{R'} = \left(\frac{100 - \ell}{\ell} \right)$$

$$\frac{100 - \ell}{\ell} = \frac{X}{2R} = \frac{3}{2}$$

$$\ell = 40.00 \text{ cm}$$



Q26



Both diodes are in reverse biased

$$I = \frac{9}{30} = \frac{3}{10} \text{ A} = 0.3 \text{ A}$$

Hints & Solutions

Physics